

Development of Respiratory System

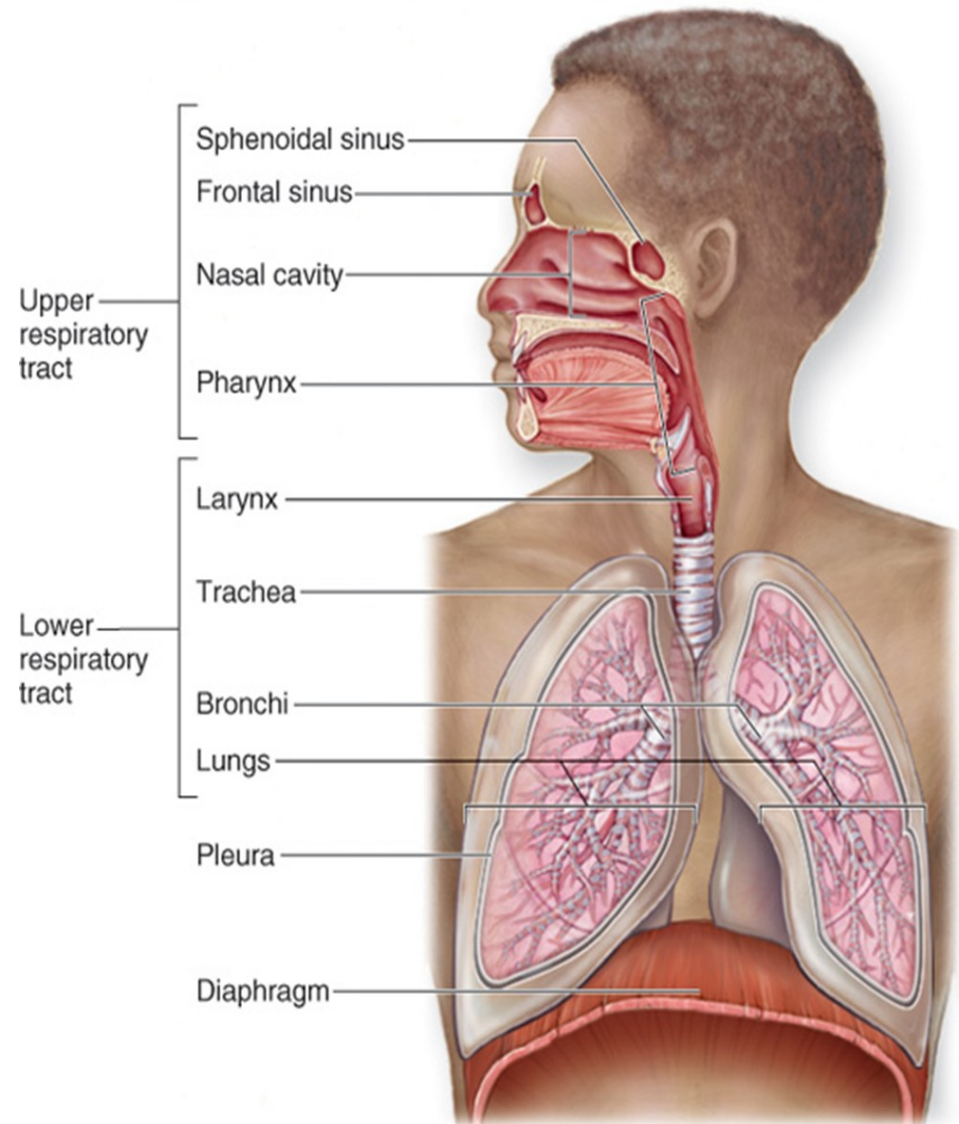
Respiratory System

- **Upper respiratory tract:**

- Nose
- Nasal cavity & paranasal sinuses
- Pharynx

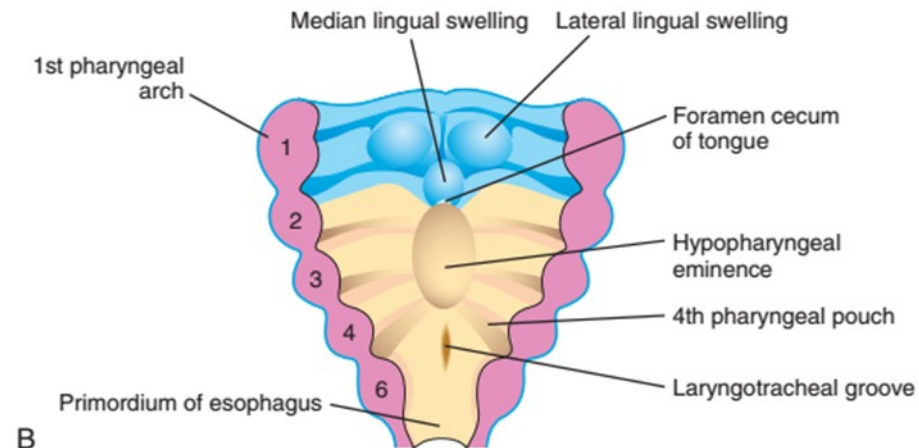
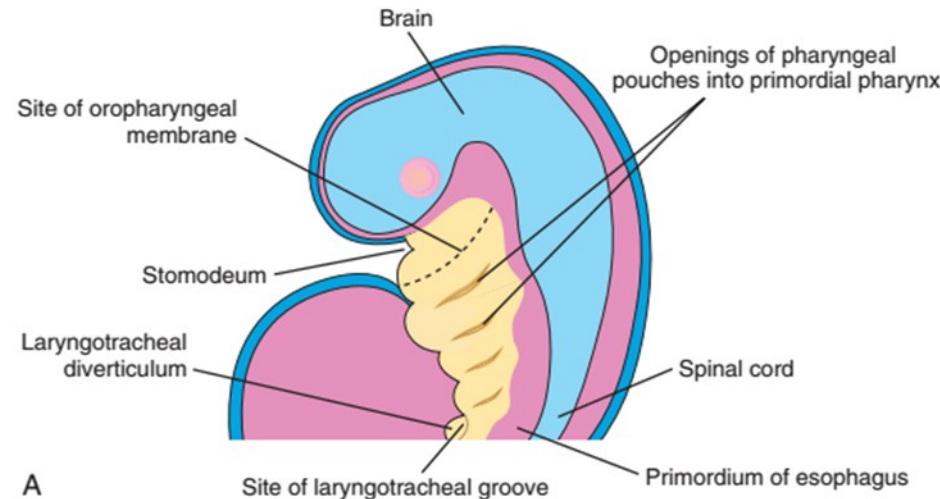
- **Lower respiratory tract:**

- Larynx
- Trachea
- Bronchi
- Lungs



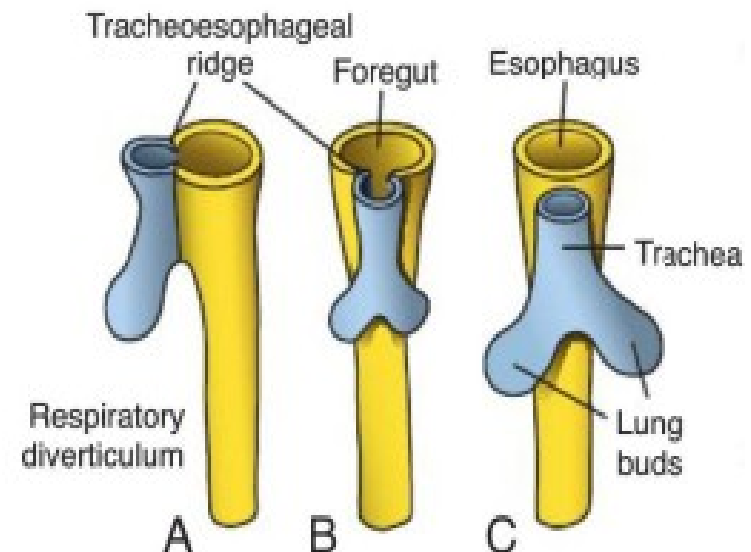
Development of Lower Respiratory Tract

- develops from the **respiratory diverticulum** (lung bud).
- Begins to form during the **4th** week of development
- Begins as a midline outgrowth (**laryngotracheal groove**) from the caudal part of the ventral wall of the primitive pharynx (foregut) just caudal to 4th pharyngeal pouch
- The groove envaginates and forms the **laryngotracheal (respiratory) diverticulum**

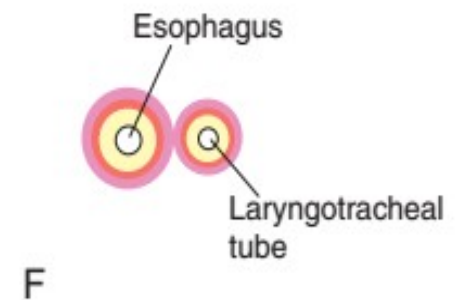
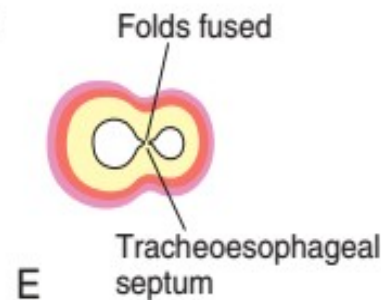
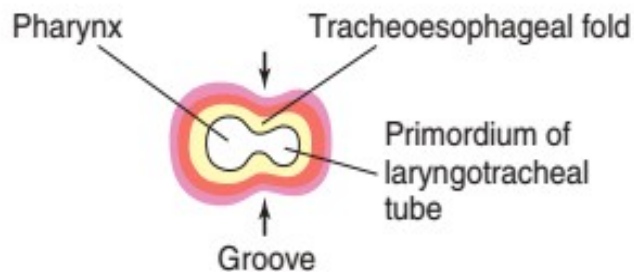
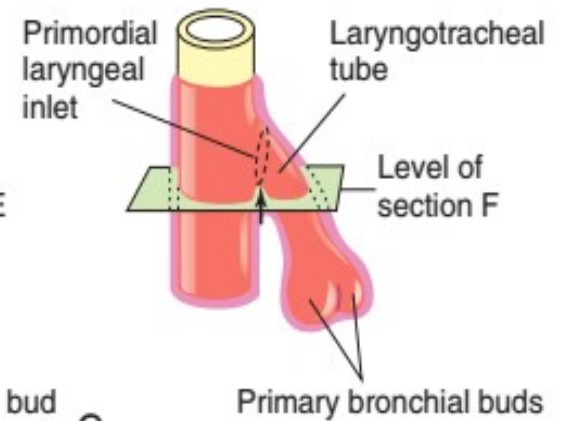
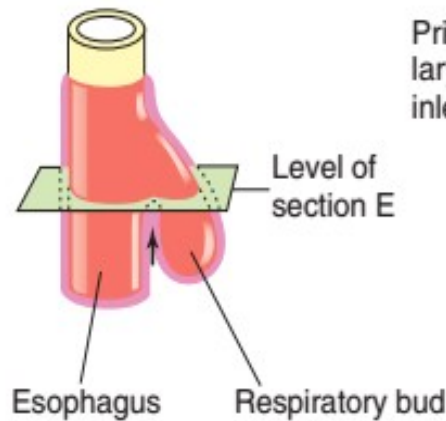
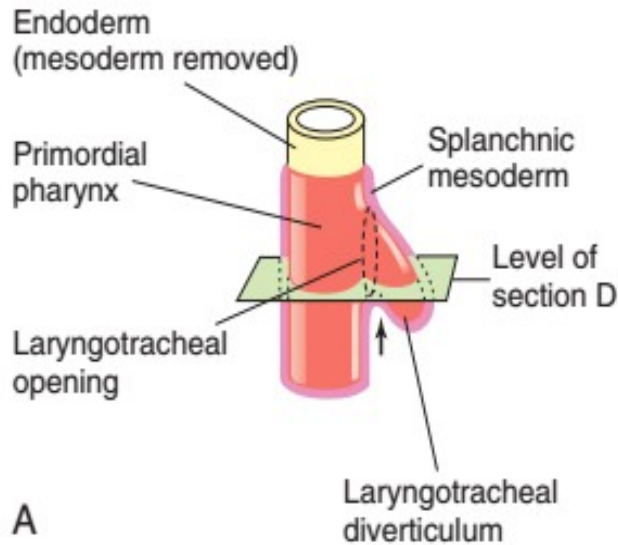


Development of Lower Respiratory Tract

- Initially the diverticulum is in open communication with the foregut.
- When the diverticulum expands caudally, two longitudinal ridges known as the **tracheoesophageal ridges (folds)**
 - separate the developing diverticulum from the foregut.
- These ridges fuse to form the **tracheoesophageal septum**.
- Tracheoesophageal septum** divides the foregut into a:
 - Dorsal portion:** primordium of the oropharynx and esophagus
 - Ventral portion: laryngotracheal tube** (primordium of the larynx, trachea, bronchi, and lungs)

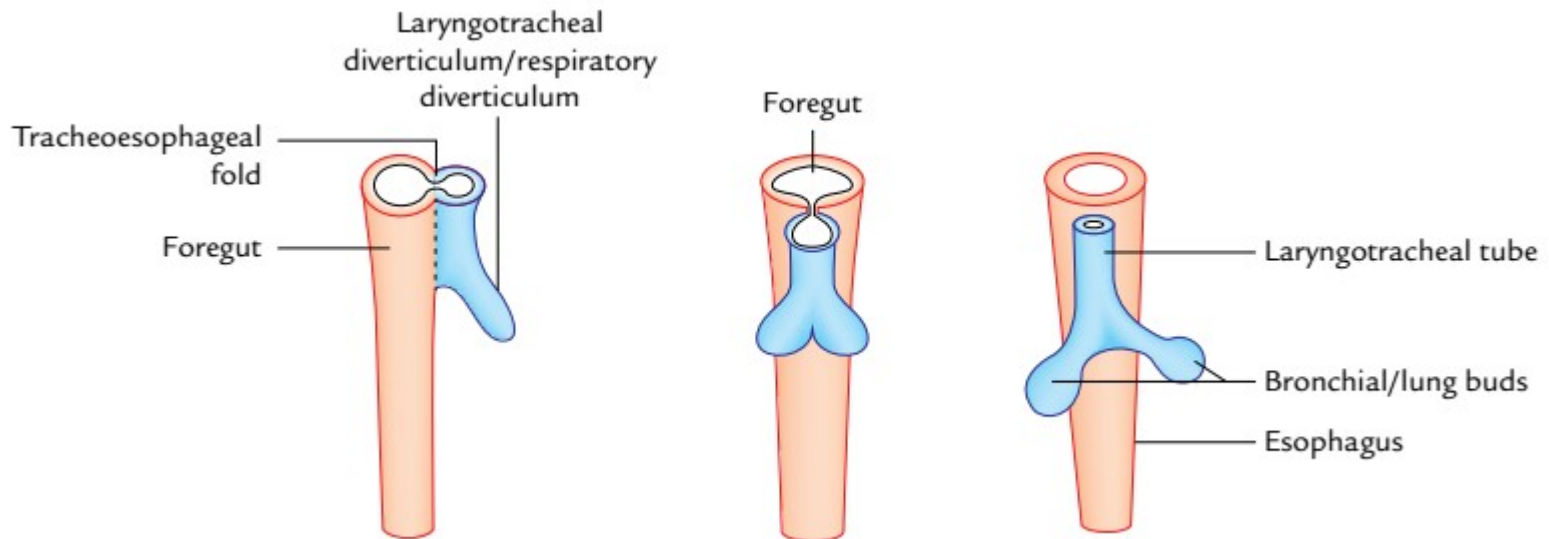


Development of Lower Respiratory Tract



Development of Lower RT

- The respiratory primordium maintains its communication with the pharynx through the **laryngeal inlet**
- **proximal part** of the respiratory diverticulum remains **tubular** and forms **larynx & trachea**.
- **distal end** of the diverticulum **dilates** to form **lung bud**, which divides to give rise to 2 lung buds (primary bronchial buds)

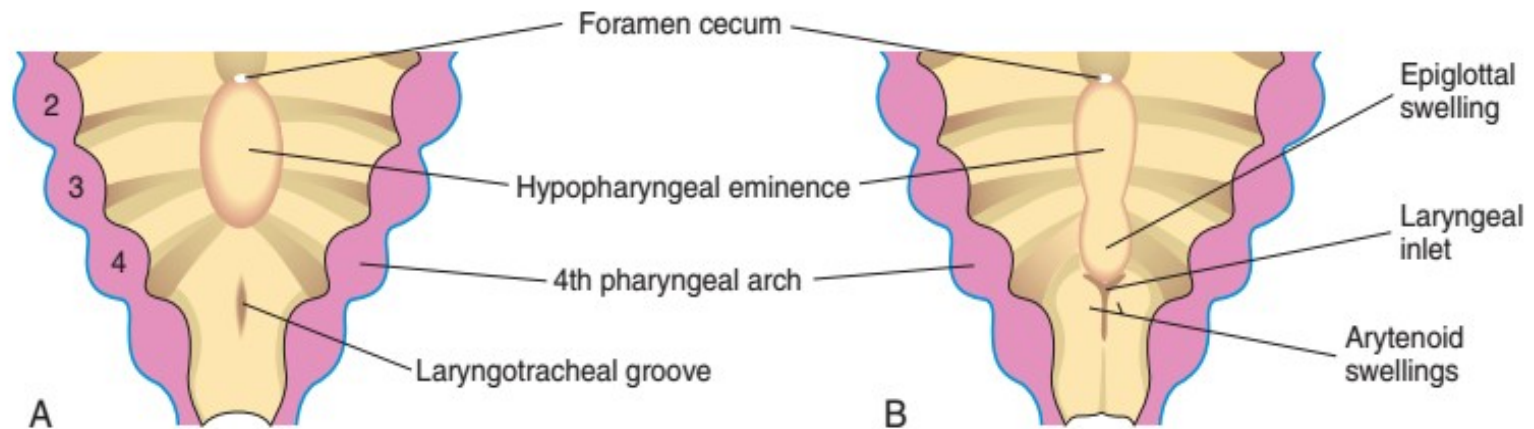


Development of Lower RT

- The **endoderm lining** the laryngotracheal diverticulum gives rise to the:
 - Epithelium & Glands of the respiratory tract
- The surrounding **splanchnic mesoderm**, derived from neural crest cells, gives rise to the:
 - Connective tissue, **Cartilage** & **Smooth muscles** of the respiratory tract

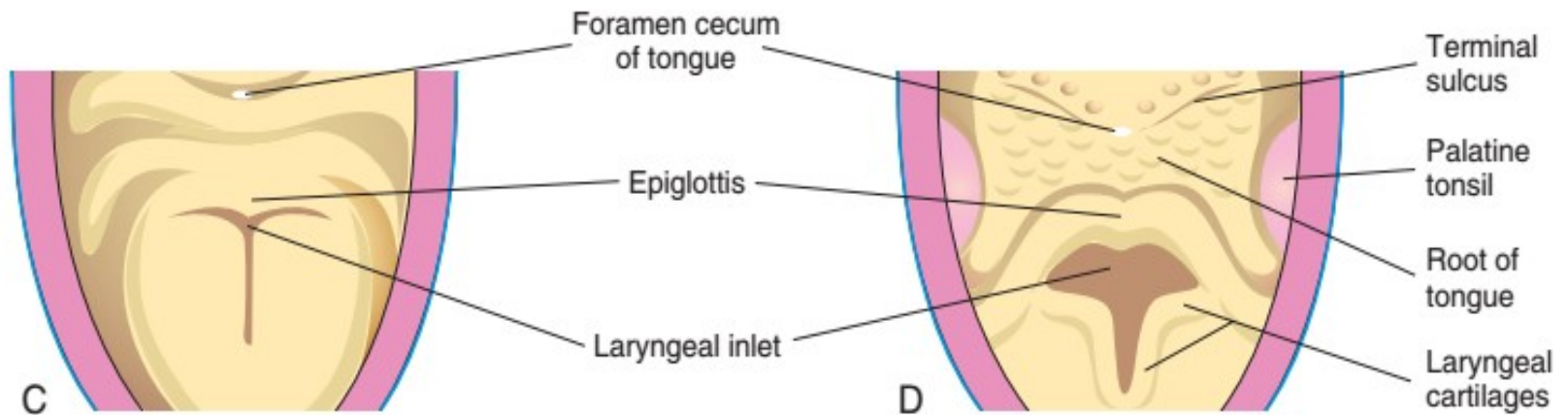
Development of Larynx

- develops from the **cranial-most part of laryngotracheal tube**
- The communication between the laryngotracheal diverticulum and primitive pharynx persists as a **laryngeal inlet**.
- The **epithelium & glands** are derived from **endoderm**.
- Laryngeal **muscles & the cartilages** of the larynx **except epiglottis**, develop from the **mesoderm of 4th & 6th pairs of pharyngeal arches**, which is derived from neural crest cells.



Development of Larynx

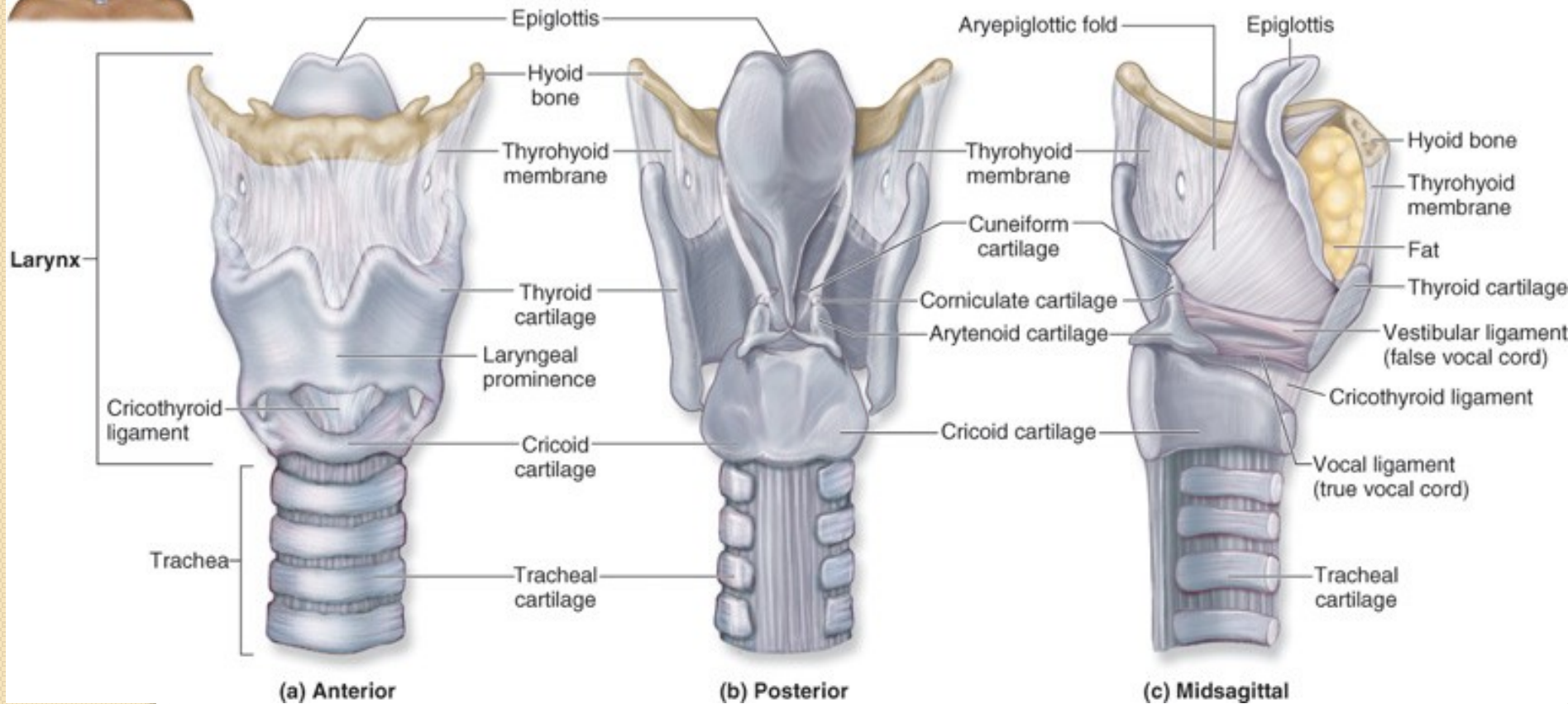
- The mesenchyme (of 4th and 6th pharyngeal arches) surrounding the laryngeal orifice proliferates.
- As a result, the slit-like laryngeal orifice becomes **T-shaped**
- Subsequently, when mesenchyme of the two arches transforms into the **thyroid**, **cricoid**, and **arytenoid cartilages**, the characteristic adult shape of the laryngeal orifice can be recognized.



Recanalization of Larynx

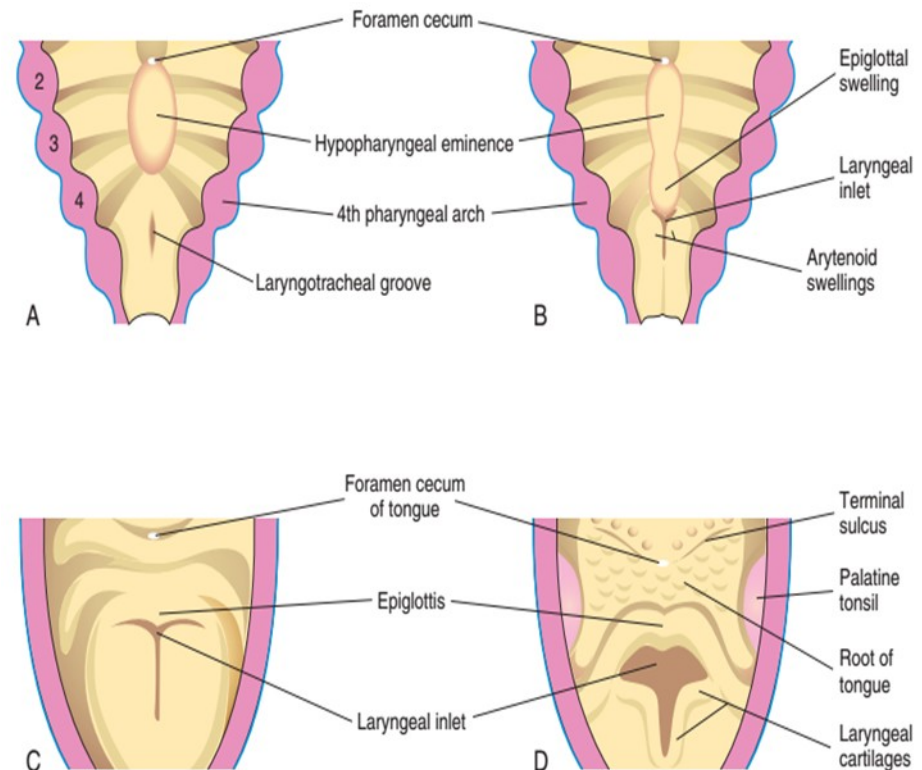
- The laryngeal epithelium proliferates rapidly resulting in temporary occlusion of the laryngeal lumen
- Later on the cells obliterating the lumen breakdown and recanalization of larynx takes place by the **10th week**.
- During recanalization, endodermal cells form two pairs of folds:
 - **upper pair** of vestibular folds give rise to **false vocal cords**
 - **lower pair** of vocal folds give rise to **true vocal cords**
 - A pair of lateral recesses bound by these folds is called **laryngeal ventricles**.

Recanalization of Larynx



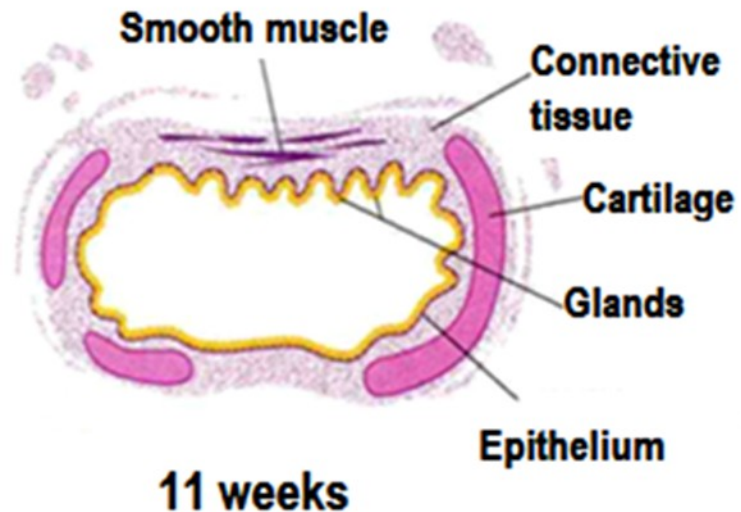
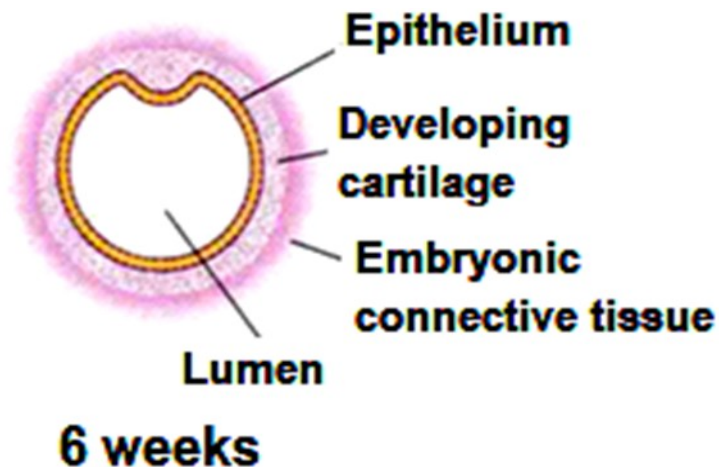
Epiglottis

- It develops from the caudal part of the **hypopharyngeal eminence**, a swelling formed by the proliferation of mesoderm in the ventral ends of the **third and fourth** pharyngeal arches.
- Growth of the larynx and epiglottis is rapid during the first three years after birth. By this time the epiglottis has reached its adult form.

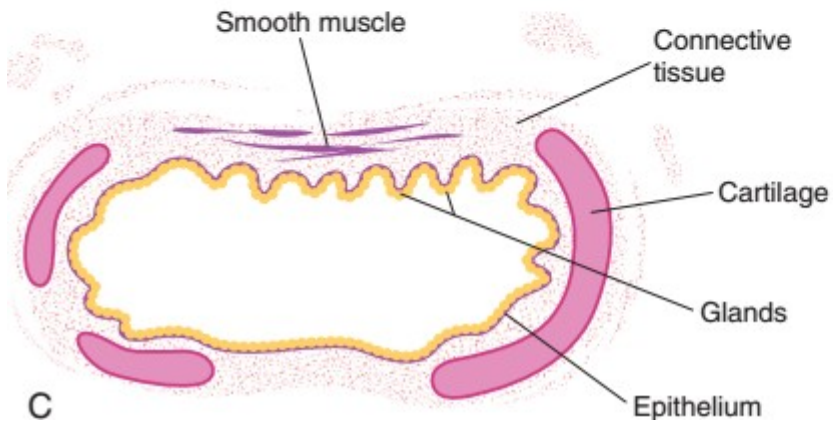
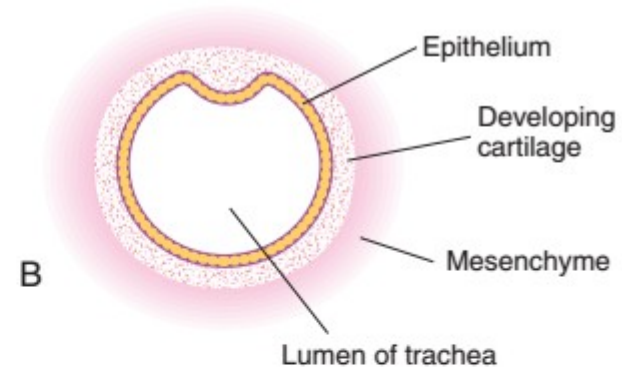
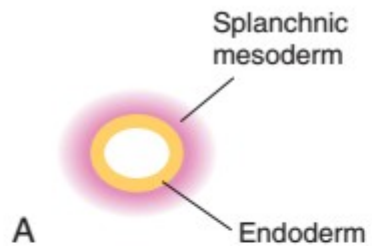


Development of the Trachea

- develops from part of the **laryngotracheal tube** (respiratory diverticulum), which lies between the larynx and its bifurcation
- The **endoderm** of laryngotracheal diverticulum **forms** the **lining epithelium and glands** of the trachea.
- The cartilages, connective tissue, and muscles of the trachea are derived from the surrounding **splanchnopleuric mesoderm**.

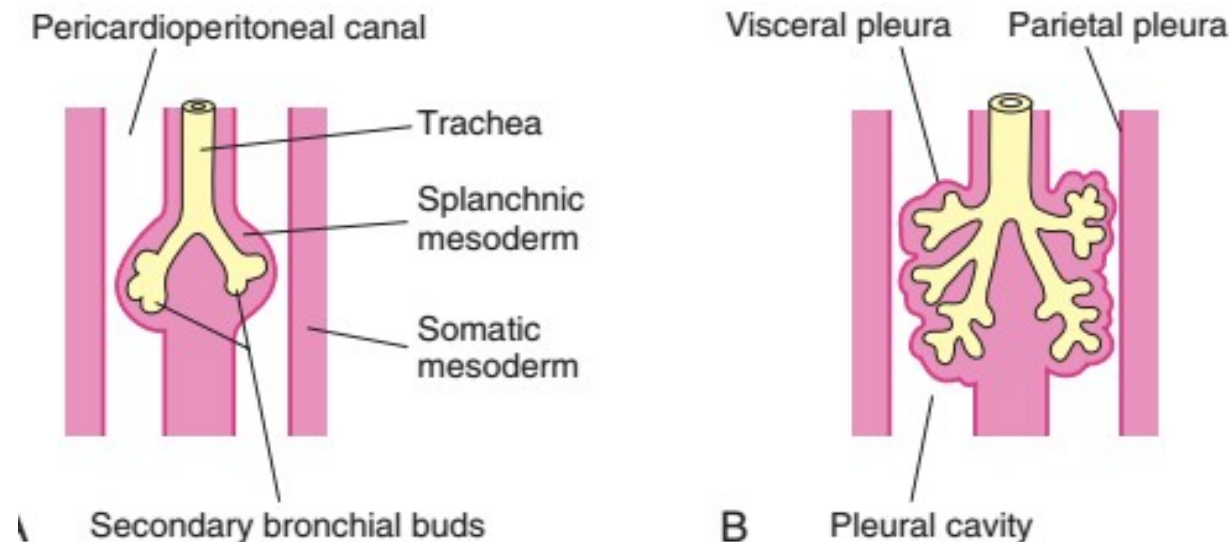


Development of the Trachea



Development of the Bronchi & Lungs

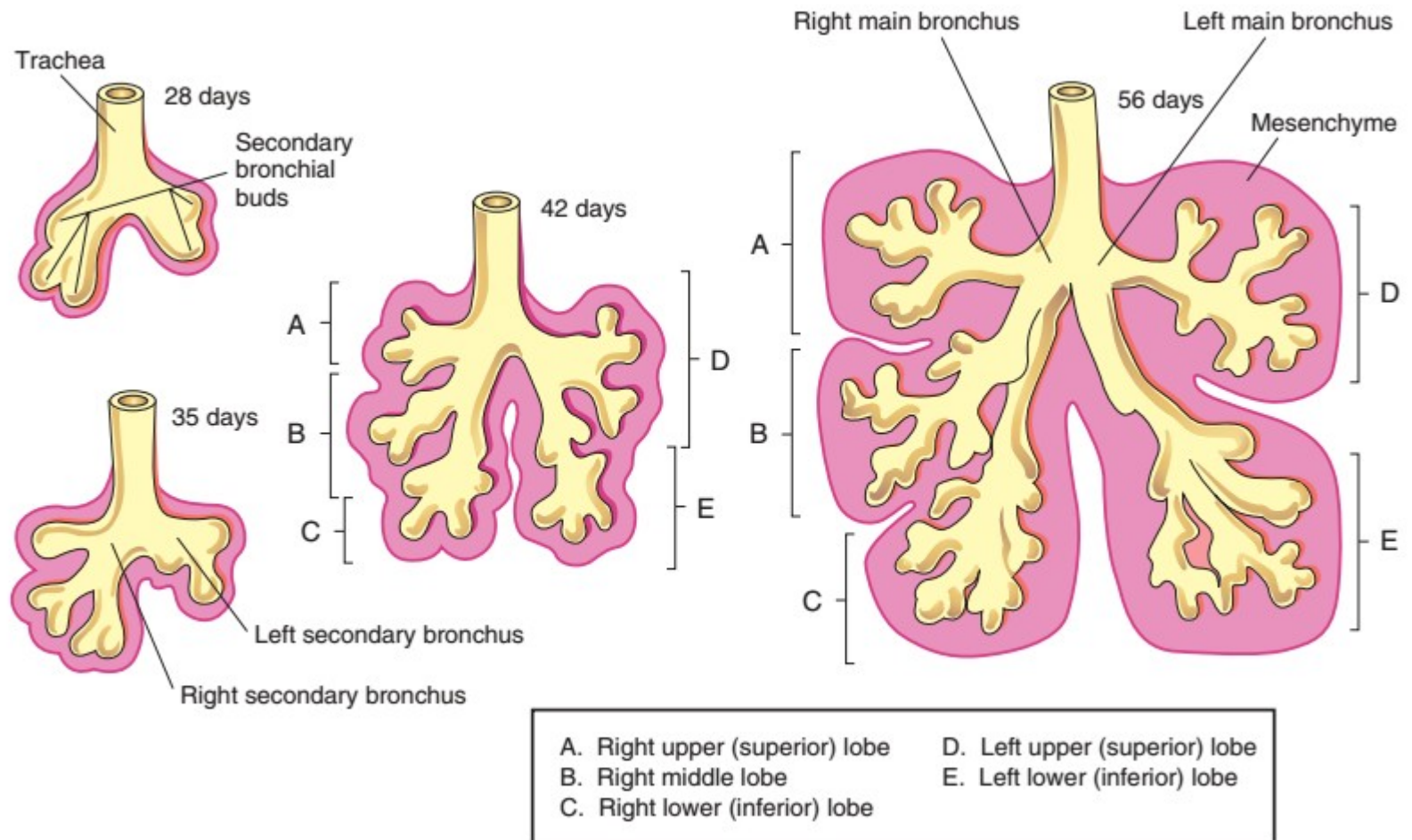
- The laryngotracheal (respiratory) diverticulum divides into **two bronchial buds**.
- The 2 primary bronchial buds grow laterally into the **pericardioperitoneal canals** (part of the intraembryonic celom), the primordia of pleural cavities
- At the beginning of the **5th week**, each bronchial bud develops into a **principal (main) bronchus**.



Development of the Bronchi & Lungs

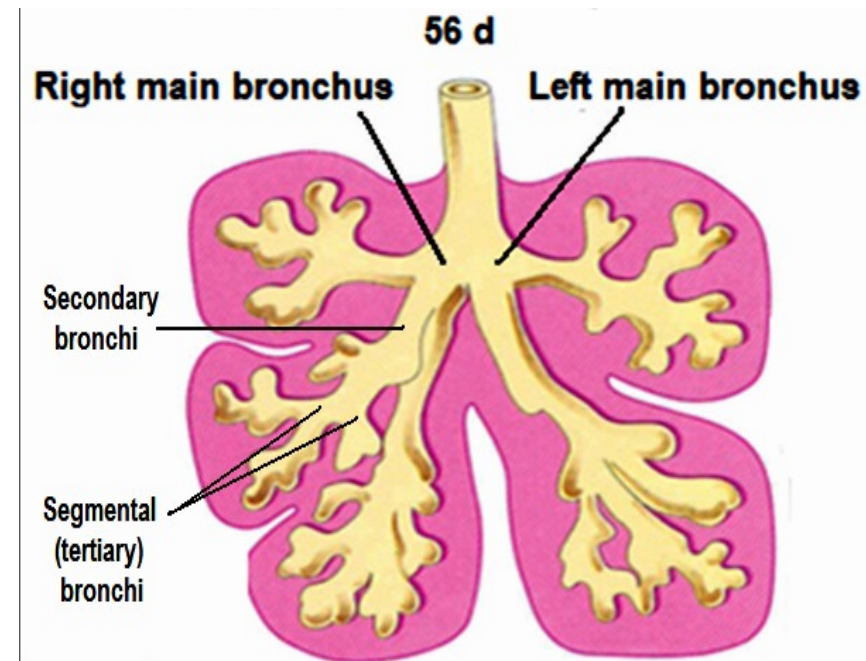
- The **right main bronchus** is slightly **larger** than the left and is oriented **more vertically**.
- The left main bronchus lies more transversely than the right
- The main bronchi subdivide into **secondary bronchi** that form **lobar**, and **segmental** branches.
- **On the right**
 - **superior lobar bronchus** will supply the **upper lobe** of the lung
 - **inferior bronchus** subdivides into **two bronchi**, one to the middle lobe of the right lung and the other to the lower lobe.
- **On the left**, the two secondary bronchi supply the upper and lower lobes of the lung.
- Each lobar bronchus undergoes progressive branching.

Development of the Bronchi & Lungs



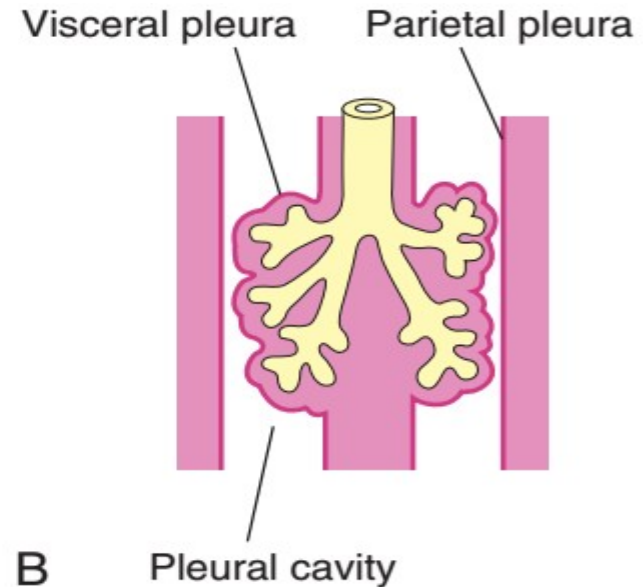
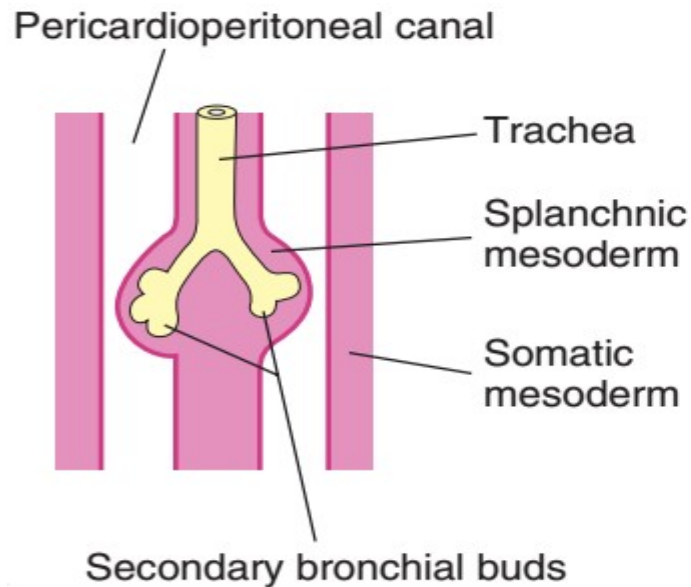
Development of the Bronchi & Lungs

- The **segmental bronchi** begin to form by the **7th week**.
 - 10 in the right lung and 8 or 9 in the left lung,
- The surrounding mesenchyme also divides.
- The segmental bronchi, with the surrounding mass of mesenchyme, form the primordia of the **bronchopulmonary segments**.
- By 24 weeks, 17 orders of branches have formed & **respiratory bronchioles** have developed.
- An additional seven orders of airways develop after birth.



Development of the Bronchi & Lungs

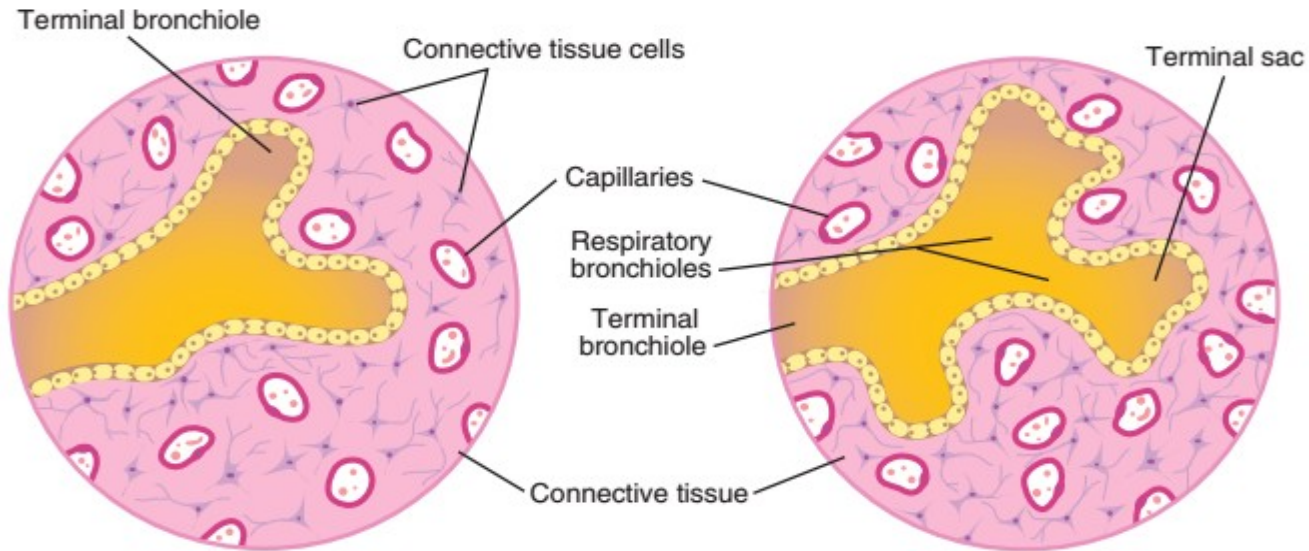
- As the bronchial buds grow laterally into the **pericardioperitoneal canals**, the primordia of the pleural cavities
 - The **splanchnic mesoderm** develops into the **visceral pleura**.
 - The **somatic mesoderm** layer becomes the **parietal pleura**.
 - The space between the parietal and visceral pleura forms the **pleural cavity**.



Maturation of Lungs

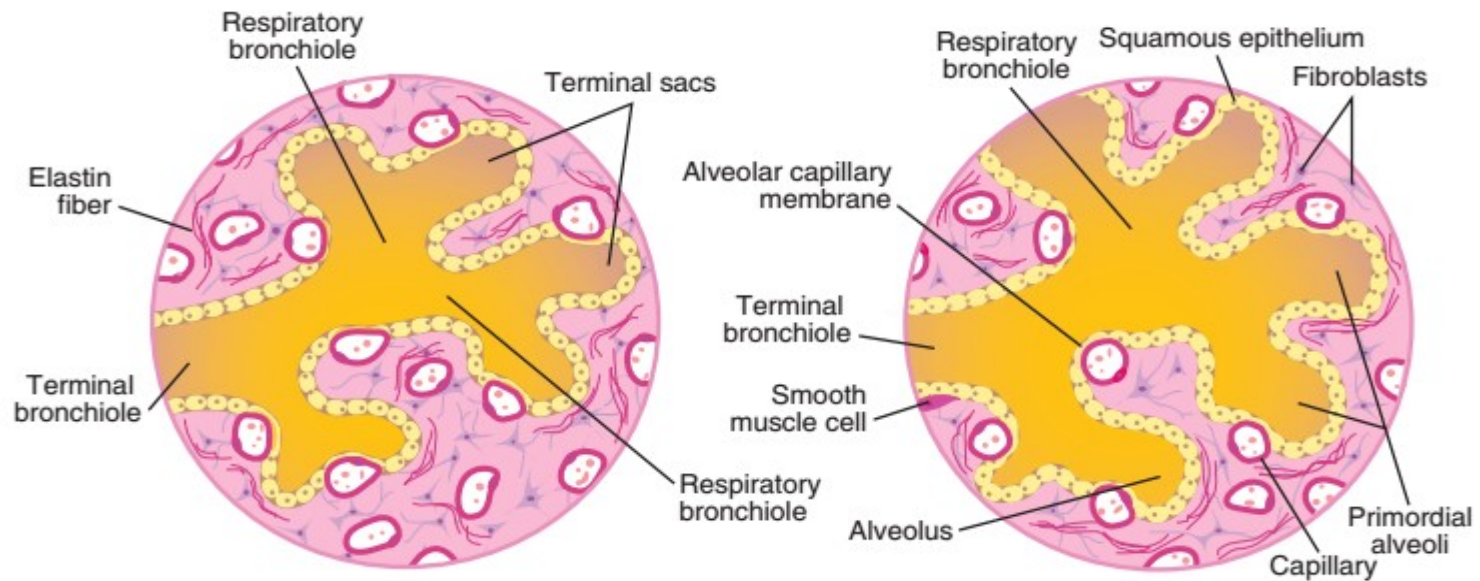
- Maturation of lung is divided into 4 periods:
 - **Pseudoglandular Stage** (5 - 17 weeks)
 - **Canalicular Stage** (16 - 25 weeks)
 - **Terminal sac Stage** (24 weeks - birth)
 - **Alveolar Stage** (late fetal period – 8 years)
- These periods overlap each other because the cranial segments of the lungs mature faster than the caudal ones.

Maturation of Lungs



A Pseudoglandular period (6–16 weeks)

B Canalicular period (16–26 weeks)

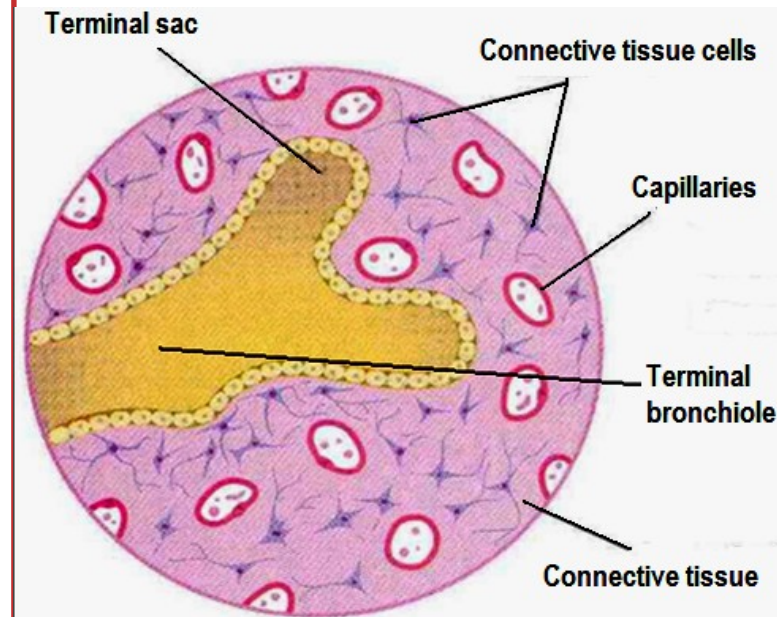


C Terminal saccular period (26 weeks–birth)

D Alveolar period (32 weeks–8 years)

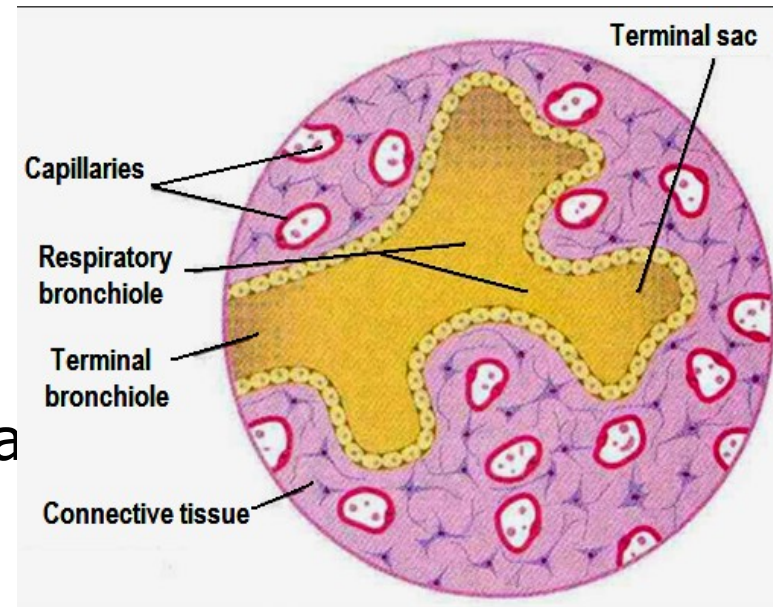
Pseudoglandular Stage (5 - 17 weeks)

- Developing lungs somewhat resembles an **exocrine gland**
- By 16 weeks, Bronchial tree is formed up to **terminal bronchioles**
- **Respiratory portion** of the bronchial tree (e.g., respiratory bronchioles, alveolar ducts, and alveoli) **are NOT formed**
- Respiration is NOT possible at this stage
- Fetuses born during this period are unable to survive.



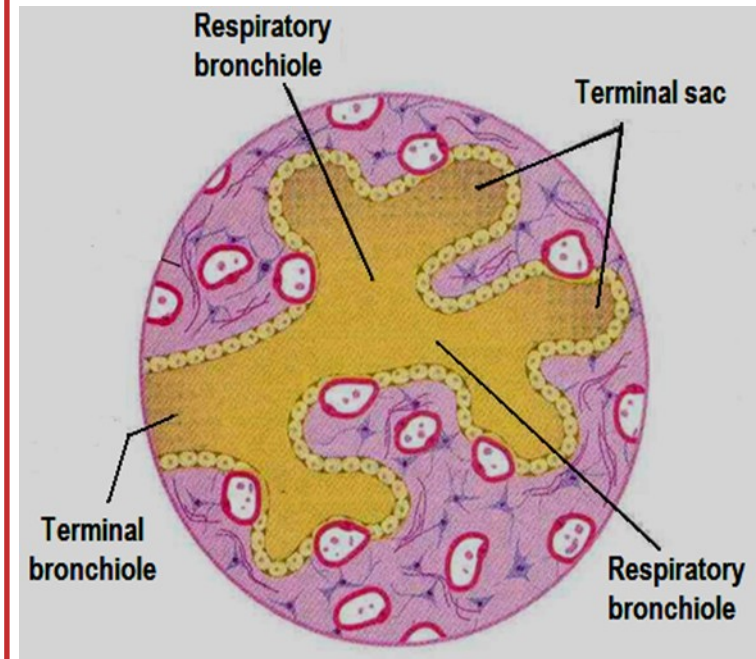
Canalicular Stage (16 - 25 weeks)

- By 24 weeks each terminal bronchiole has given rise to two or more respiratory bronchioles.
- The respiratory bronchioles divide into 3 to 6 tubular passages called alveolar ducts.
- Some thin-walled **terminal sacs** (primordial alveoli) develop at the end of respiratory bronchioles.
- **Lumina** of bronchi and terminal bronchioles **become larger**.
- Lung tissue is well vascularized
- Respiration is **POSSIBLE** at the end of this period.
- **Fetus born** at the end of this period may survive if given intensive care (but **usually die**)



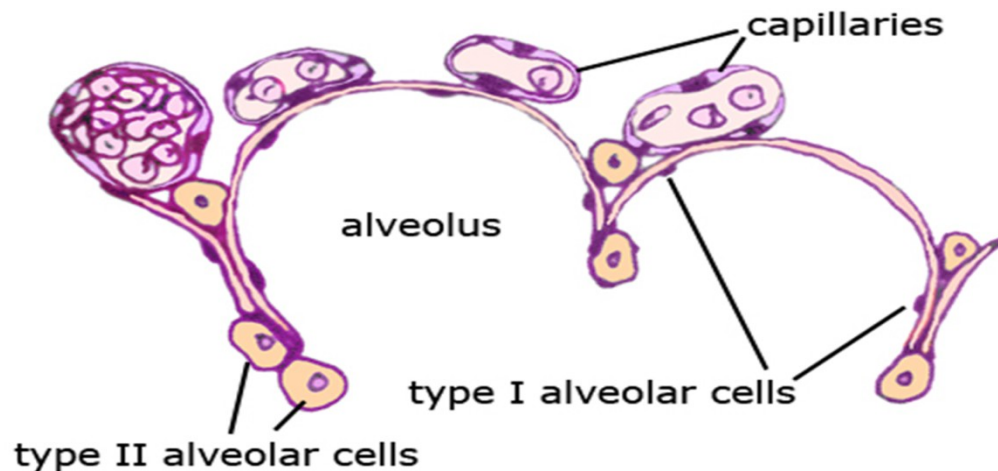
Terminal Sac Period (24 weeks - birth)

- Many **more terminal sacs develop**.
- Their **epithelium** becomes thin as it **changes** cell shape from columnar to **squamous**
- Capillaries begin to bulge into developing alveoli.
- **Adequate gas exchange can occur** which allows the prematurely born **fetus to survive**



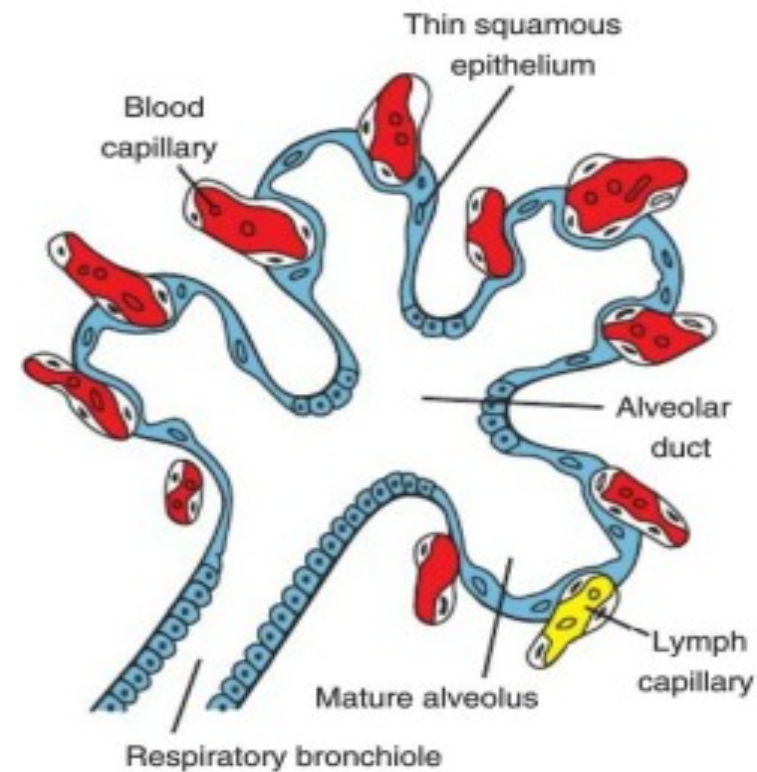
Terminal Sac Period (24 weeks - birth)

- By 24 weeks, the terminal sacs are lined by:
 - Squamous **type I pneumocytes** which are endodermal in origin
 - Rounded secretory epithelial cells, **type II pneumocytes**, that secrete a mixture of phospholipids called **surfactant** (lowers surface tension at the air-alveolar interface).
- This intimate contact between epithelial of the alveoli and the endothelial cells of the capillaries forms the **blood-air barrier**.



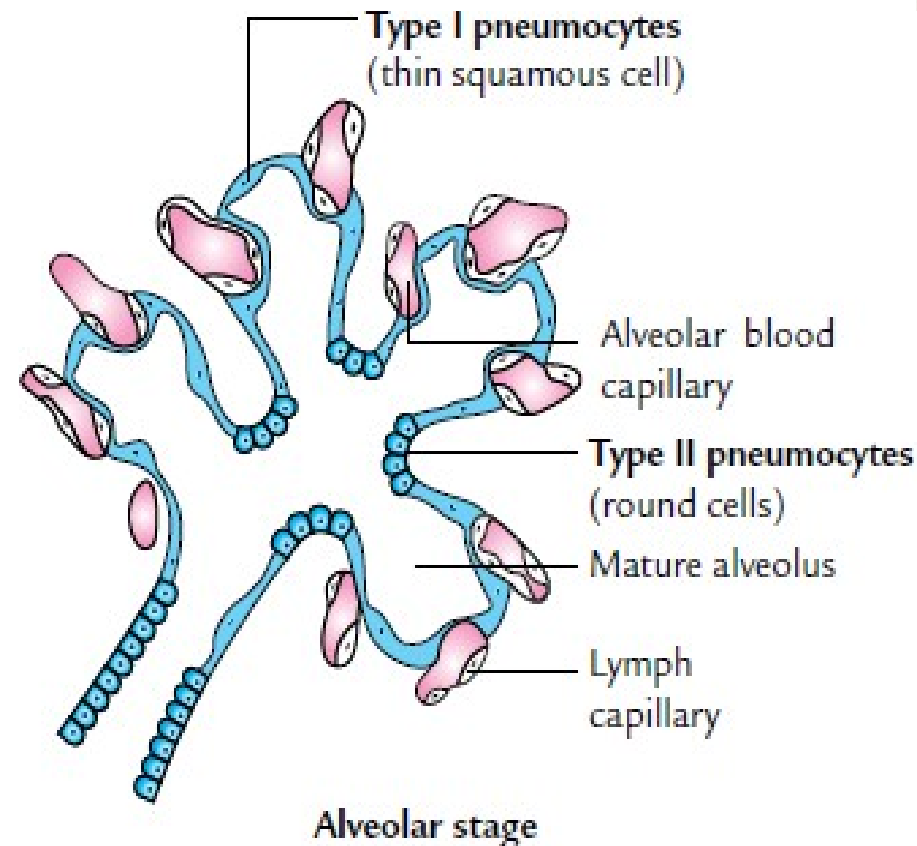
Alveolar Stage (late fetal period – 8 years)

- At the beginning of the alveolar period, each respiratory bronchiole terminates in a cluster of thin-walled **terminal sacculles**, separated from one another by loose connective tissue.
 - These terminal sacculles represent future **alveolar sacs**.
- The epithelial lining (Type I pneumocytes) of the terminal sacs attenuates to an **extremely thin squamous epithelial layer**.
 - for gaseous exchange.



Alveolar Stage (late fetal period – 8 years)

- Type II pneumocytes **produce** a sufficient amount of **surfactant**, particularly during the last 2 weeks before birth.
- Mature alveoli is not formed until after birth, **95% of mature alveoli develops after birth.**
- About **50 million alveoli**, one sixth of the adult number of alveoli are **present at birth.**



Alveolar Stage (late fetal period – 8 years)

- The **lungs increase in size** due to
 - an **increase in the number of** respiratory bronchioles and primordial alveoli
 - NOT from an increase in the size of the alveoli.
- From 3-8 year or so, **the number of immature alveoli continues to increase**. Unlike mature alveoli, immature alveoli have the potential for forming additional primordial alveoli.
- By about the **eighth year**, the adult complement of **300 million alveoli is present**.

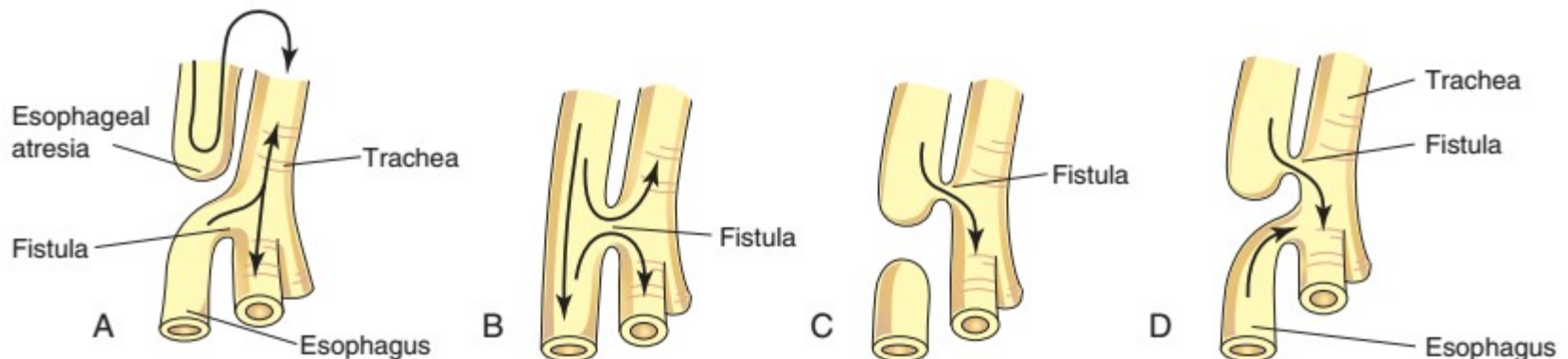
Clinical: Developmental anomalies of LRT

- **Laryngeal atresia**

- **Failure of recanalization** of the larynx, which produces obstruction of the upper fetal airway

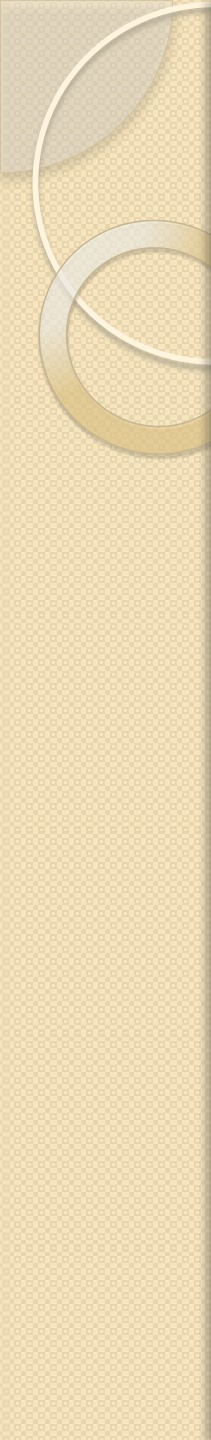
- **Tracheoesophageal fistula**

- Abnormal passage between the trachea and esophagus
- results from **incomplete division of the cranial part** of the **foregut** into respiratory and esophageal parts during the 4th week
- It is generally associated with **esophageal atresia**, which will then result in **polyhydramnios**



Clinical: Developmental anomalies of LRT

- Tracheal stenosis & atresia
- Congenital lung cysts
- Agenesis of lungs
- Lung hypoplasia
- Accessory lungs



Thank You !!!